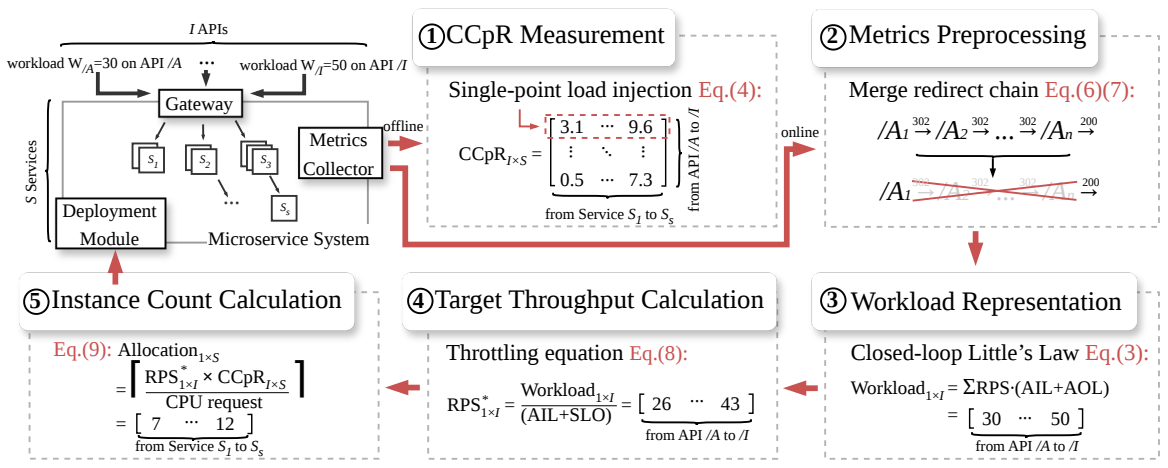

I APIs
workload W/A=30 on API /A
workload W/I=50 on API /I
Gateway
S Services
Deployment Module
Microservice System
Metrics Collector
offline
online
①CCpR Measurement
Single-point load injection Eq.(4):
CCpRI×S =
3.1 ⋯ 9.6
0.5 ⋯ 7.3
from Service S1 to Ss
from API /A to /I
②Metrics Preprocessing
Merge redirect chain Eq.(6)(7):
/A1 → /A2 → ... → /An →
/A1 →
302
200
③Workload Representation
Closed-loop Little's Law Eq.(3):
WorkloadI×1 = ΣRPS·(AIL+AOL)
= 30 ⋯ 50
from API /A to /I
④Target Throughput Calculation
Throttling equation Eq.(8):
RPS*I×1 = Workload1×I / (AIL+SLO) = 26 ⋯ 43
from API /A to /I
⑤Instance Count Calculation
Eq.(9): Allocation1×S
= ⌈ RPS*I×1 × CCpRI×S / CPU request ⌉
= 7 ⋯ 12
from Service S1 to Ss